DES-ESOD: DUAL-EVIDENCE SELECTION AND COVERAGE OPTIMIZATION FOR HIGH-RESOLUTION SMALL OBJECT DETECTION

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ABSTRACT

High-resolution aerial and maritime imagery presents severe computational challenges for small-object detection because background regions heavily dominate the visual field. While selective-computation detectors significantly reduce redundant processing by dynamically routing features to prospective foreground areas, existing spatial selectors rely strictly on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently degrade toward background levels; discarding a region at this early stage results in an irrecoverable false negative for all subsequent detector stages. In this paper, we propose DES-ESOD, an efficient dual-evidence selection framework designed for recall-safe, high-resolution small object detection. DES-ESOD introduces a dual-evidence priority head that synergistically fuses semantic objectness with channel-pooled spectral saliency at negligible compute overhead ($\mathcal{O}(1)$ relative to backbone width), capturing critical high-frequency structural cues from semantically weak targets. Furthermore, we formulate an object-level soft-coverage loss that directly optimizes the joint spatial coverage probability of all ground-truth instances, preventing early false dismissals. Paired with a lightweight ISPP-based decoupled head and scale-aware Wasserstein regression, DES-ESOD demonstrates consistent performance gains across the UAVDT and SeaPerson benchmarks, achieving +2.7% mAP and +5.5% total detector recall on UAVDT while preserving real-time inference capability.

Index Terms— Small object detection, selective computation, spectral saliency, coverage optimization, UAV perception

1. INTRODUCTION

High-resolution visual sensing from unmanned aerial vehicles (UAVs) and remote platforms is essential for applications such as traffic monitoring, precision agriculture, and search-and-rescue. However, detecting small and tiny targets in high-resolution imagery poses significant computational and perceptual challenges [1, 2]. In these scenarios, objects of interest typically span only a few pixels (often less than 16×16 pixels), while background clutter occupies over 70% to 80% of the spatial area [3, 4]. Running modern dense object detectors on full-resolution feature maps incurs prohibitive arithmetic cost and memory overhead, making onboard real-time deployment on resource-constrained edge devices extremely difficult.

To eliminate redundant computations on uniform background regions, *selective computation* has emerged as an effective paradigm [3, 5–7]. Methods such as QueryDet [6] and CEASC [7] perform cascaded sparse querying or adaptive activation masking, while ESOD [3] predicts an early-stage spatial heatmap on shallow features to route only candidate foreground patches to deeper backbone, neck, and detection heads.

Despite their computational efficiency, existing spatial selectors share a fundamental limitation: they rely almost exclusively on a single semantic foreground or objectness response. While effective for medium and large objects, this assumption breaks down severely for genuinely tiny targets. Under spatial downsampling, the feature footprint of a tiny object collapses into only one or two activations that frequently fade into surrounding background textures [8]. More critically, spatial selection decisions exhibit a severe *asymmetric cost*: retaining an extra background patch only marginally increases computational budget, whereas prematurely discarding a patch containing a true tiny object causes a catastrophic, unrecoverable false negative for all subsequent detector stages [3].

To overcome this bottleneck, we explore whether an orthogonal evidence source can complement semantic representations during early routing. Spectral analysis indicates that tiny objects retain distinctive high-frequency structural gradients and spectral saliency even when their semantic activations are weak [8]. Furthermore, we observe that conventional per-pixel binary cross-entropy (BCE) loss treats candidate cells independently, often failing to guarantee that at least one candidate cell actually covers a ground-truth object.

In this work, we propose **DES-ESOD** (Dual-Evidence Selection for Efficient Small Object Detection), a novel framework built upon the principles of multi-modal evidence fusion and coverage-constrained spatial routing. Specifically, DES-ESOD constructs a dual-evidence priority head that combines semantic objectness with an ultra-lightweight channel-pooled spectral saliency branch, utilizing trainable Laplacian and Sobel filter banks at negligible computational overhead ($\mathcal{O}(1)$

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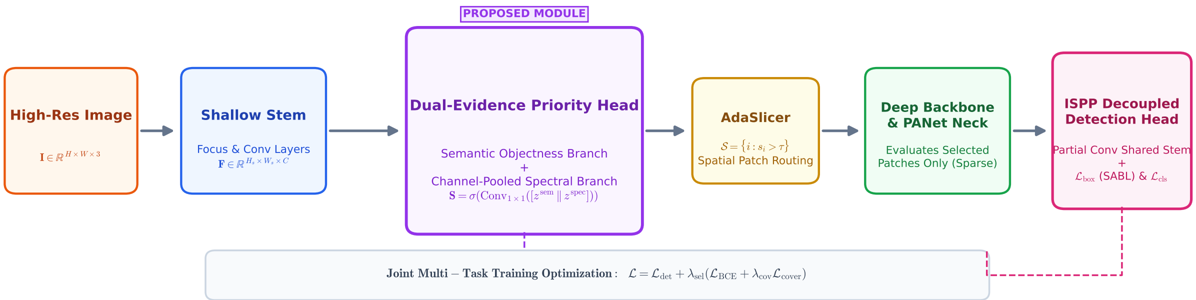

Fig. 1. System architecture of DES-ESOD. A high-resolution input image is processed through a shallow feature stem. The proposed dual-evidence priority head (§3.1) synergistically fuses semantic objectness with channel-pooled spectral saliency. Dynamic patch extraction routes candidate regions containing high priority to the downstream backbone and lightweight ISPP decoupled detection head, optimized via the object-level soft-coverage loss and scale-aware balancing loss (§3.2, §3.3).

relative to backbone channel width). To guarantee recall safety, we design an object-level soft-coverage loss ($\mathcal{L}_{\mathrm{cover}}$) that directly optimizes the joint spatial coverage probability for every ground-truth bounding box. Finally, we incorporate an inverted residual partial-convolution (ISPP) decoupled head and a scale-aware balancing loss (SABL) [9] to simultaneously reduce head computation and enhance micro-scale bounding-box localization.

The main contributions of this paper are summarized as follows:

1. We identify the fundamental vulnerability of single-evidence semantic routing in selective computation and propose a dual-evidence priority head that synergistically fuses semantic and spectral saliency at negligible arithmetic overhead.

2. We formulate an object-level soft-coverage loss ($\mathcal{L}_{\mathrm{cover}}$) that directly optimizes the multi-cell joint coverage probability across all ground-truth instances, preventing premature spatial discarding of tiny objects.

3. We introduce an efficient ISPP-based decoupled detection head integrated with scale-aware Wasserstein balancing loss (SABL), boosting tiny-object localization while maintaining low latency.

4. Extensive evaluations on the challenging UAVDT and SeaPerson benchmarks demonstrate that DES-ESOD achieves substantial improvements in tiny-object recall and mAP over state-of-the-art selective computation baselines.

2. RELATED WORK

Small Object Detection in Aerial Imagery. Detecting small and tiny targets in aerial and maritime imagery is a long-standing challenge due to extreme scale variations, low contrast, and complex background textures [1, 2, 4, 10]. Early approaches often utilize image-level cropping or density-guided patch generation (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]). While these methods reduce the area processed at high resolution, they typically rely on separate coarse detector stages or incur repeated feature extraction on overlapping crops. Other works focus on specialized label assignment strategies and geometric loss functions for micro-scale bounding boxes, such as Normalized Gaussian Wasserstein Distance (NWD) [14], RFLA [15], and scale-aware balancing networks (SSABNet) [9]. However, these dense detection frameworks still execute full-resolution convolutions uniformly across entire images, failing to eliminate massive spatial computation on empty background areas.

Selective and Dynamic Computation. To mitigate arithmetic redundancy, selective computation dynamically routes computations to spatial regions containing prospective targets. AutoFocus [5] predicts focus pixels on coarse scales to guide fine-scale processing. QueryDet [6] accelerates high-resolution feature pyramids via cascaded sparse queries, while CEASC [7] formulates adaptive activation masking (AMM) to skip background locations in sparse convolutions. ESOD [3] advances this paradigm by integrating an early ObjSeeker module right after the shallow network stem, slicing features into patches (AdaSlicer) and bypassing background regions across all subsequent backbone, neck, and detection head stages. Nonetheless, existing selectors share two critical limitations: (i) they rely exclusively on single semantic objectness scores, which easily collapse for tiny objects under downsampling; and (ii) they are supervised via per-cell classification losses that offer no object-level coverage guarantees for micro instances.

Spectral Analysis and Spatial Saliency. Frequency-domain methods provide orthogonal insights into visual rep-

Table 1. Systematic comparison of related detection paradigms and the proposed DES-ESOD framework.

Method	Routing Granularity	Evidence Source	Coverage Guarantee	Decoupled Head Design
QueryDet [6]	Feature Pyramid / Query	Semantic only	×	Standard Conv Head
CEASC [7]	Activation Masking	Semantic only	×	Standard Conv Head
ESOD [3]	Early Spatial Patch	Semantic only	×	Standard / Sparse Head
SET [8]	Full Dense Feature Map	Spectral Filtering	×	Standard Conv Head
SSABNet [9]	Full Dense Feature Map	Semantic only	×	Dense Balancing Head
DES-ESOD (Ours)	**Early Spatial Patch**	**Dual (Semantic + Spectral)**	✓	**Lightweight ISPP + SABL**

resentations. SET [8] demonstrated that tiny objects preserve distinct high-frequency structural signatures even when their semantic features are weak, using frequency background smoothing to enhance dense feature representations. In contrast to SET, which applies dense spectral operations over entire multi-scale feature maps, DES-ESOD repurposes spectral and gradient saliency as an ultra-lightweight, channel-pooled routing signal ($\mathcal{O}(1)$ relative to backbone width), directly resolving the false-negative vulnerability in spatial selective computation.

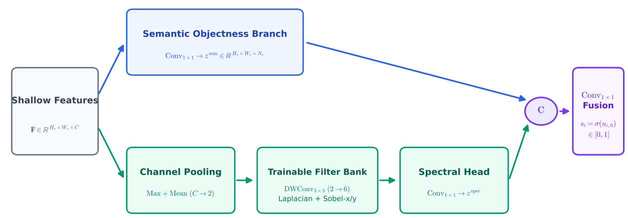

Fig. 2. Architecture of the dual-evidence priority head. The semantic branch (top) captures category-level objectness, while the channel-pooled spectral branch (bottom) extracts directional gradient and high-frequency saliency via trainable Laplacian and Sobel filter banks. The two representations are fused via a 1×1 convolution.

3. PROPOSED METHOD

The overall pipeline of DES-ESOD is illustrated in Fig. 1. Given a high-resolution input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, shallow features are extracted and passed to the proposed dual-evidence priority head (§3.1), which predicts a spatial routing priority map by fusing semantic objectness with channel-pooled spectral saliency. A dynamic patch extractor selects candidate regions with high priority. These selected regions alone are processed by the subsequent deep backbone and neck, followed by an efficient ISPP-based decoupled detection head trained under an object-level soft-coverage loss and scale-aware bounding-box regression (§3.2, §3.3).

3.1. Dual-Evidence Priority Estimation

Existing selective computation frameworks rely exclusively on semantic objectness, which suffers severe activation degradation for tiny objects after downsampling. To provide an orthogonal, recall-safe signal, DES-ESOD introduces a dual-evidence priority mechanism that combines high-level semantic confidence with low-level spectral saliency.

As depicted in Fig. 2, let $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$ denote the shallow feature representation extracted after the initial network stem, and let $f_i \in \mathbb{R}^C$ denote the feature vector at spatial location i.

Semantic Objectness Branch. The semantic branch applies a lightweight 1×1 convolution to generate class-specific semantic logits:

$$z_{i,c}^{\mathrm{sem}} = \mathrm{Conv}_{1\times1}(f_i), \quad c \in \{0, 1, \ldots, N_c - 1\}, \tag{1}$$

where N_c is the number of target categories.

Channel-Pooled Spectral Branch. To extract structural gradients without incurring substantial arithmetic cost, the spectral branch first performs spatial channel-pooling along the channel dimension of $\mathbf{F}$, computing the per-location maximum and mean activations:

$$\mathbf{F}_{\mathrm{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \,\|\, \frac{1}{C} \sum_{c=1}^{C} \mathbf{F}_{:,:,c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \tag{2}$$

where $[\cdot \,\|\, \cdot]$ denotes channel-wise concatenation. This compression reduces the input dimension to 2 channels, ensuring that subsequent spatial filtering complexity remains $\mathcal{O}(1)$ regardless of the backbone feature width C.

Next, a depthwise 3×3 convolution bank is applied over $\mathbf{F}_{\mathrm{pooled}}$. To capture isotropic edge transitions and directional structural gradients, the filter kernels are initialized with discrete Laplacian ($\mathbf{K}_{\mathrm{Lap}}$) and directional Sobel ($\mathbf{K}_{\mathrm{Sob-x}}$, $\mathbf{K}_{\mathrm{Sob-y}}$) operators:

$$\mathbf{K}_{\mathrm{Lap}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{\mathrm{Sob-x}} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad \mathbf{K}_{\mathrm{Sob-y}} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \tag{3}$$

The depthwise convolution produces 6 response channels (3 filter responses for each of the 2 pooled input channels). Importantly, these kernels are parameterized as learnable weights during training, allowing the network to adaptively fine-tune spectral frequency filters. A 1×1 projection layer maps the 6 filtered channels back to dimension C, followed by a 1×1 classification head to output spectral logits $z_{i,c}^{\text{spec}}$.

Evidence Fusion. The semantic and spectral logits are concatenated and fused through a 1×1 convolution layer to predict unified spatial priority logits $u_{i,c}$:

$$u_{i,c} = \text{Conv}_{1\times1}\left(\left[z_{i,c}^{\text{sem}} \parallel z_{i,c}^{\text{spec}}\right]\right), \quad s_i = \sigma(u_{i,0}), \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid function, and $s_i \in [0, 1]$ represents the class-agnostic routing priority score at candidate cell i. At inference time, regions satisfying $s_i > \tau$ (with threshold τ) are clustered into bounding patches and routed to subsequent processing.

3.2. Object-Level Soft-Coverage Loss

Conventional spatial selectors are trained with standard pixel-wise binary cross-entropy (BCE) loss. However, per-pixel supervision does not account for instance-level coverage: a model may predict high scores for multiple redundant cells around an easy, large object while completely dropping a tiny object whose receptive field contains only one or two weakly-activated cells.

To ensure that at least one candidate cell covers each ground-truth instance, we formulate a differentiable object-level soft-coverage loss. Let $s_i \in [0, 1]$ be the soft priority probability of candidate cell i, and let $\mathcal{N}(j)$ denote the set of selector grid cells whose spatial footprints intersect ground-truth bounding box $j \in \{1, \ldots, N_{\text{gt}}\}$. Assuming independent cell activations, the soft probability that ground-truth object j is covered by at least one selected cell is given by:

$$p_j^{\text{cover}} = 1 - \prod_{i \in \mathcal{N}(j)} (1 - s_i). \tag{5}$$

If all cells within $\mathcal{N}(j)$ have zero activation ($s_i = 0$), $p_j^{\text{cover}} = 0$; conversely, if at least one cell responds strongly ($s_i \to 1$), $p_j^{\text{cover}} \to 1$.

The object-level soft-coverage loss is formulated as:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log\left(p_j^{\text{cover}} + \epsilon\right), \tag{6}$$

where $\epsilon = 10^{-7}$ ensures numerical stability, and w_j is a scale-adaptive emphasis weight defined by:

$$w_j = \text{clip}\left(\frac{4}{a_j}, 1, 5\right), \tag{7}$$

where a_j denotes the bounding box area in selector grid cell units. This weighting assigns higher penalty to tiny targets ($a_j < 4$), ensuring that micro-scale instances receive prioritized gradient backpropagation during training.

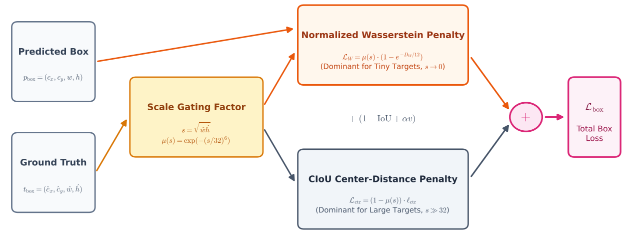

Fig. 3. Illustration of the Scale-Aware Balancing Loss (SABL). The scale gating function $\mu(s)$ adaptively balances the Normalized Gaussian Wasserstein distance for micro-scale targets and CIoU for larger objects.

The total objective function for training DES-ESOD is defined as:

$$\begin{aligned} \mathcal{L}_{\text{sel}} &= \mathcal{L}_{\text{BCE}}(u, M; \rho = 2) + \lambda_{\text{cov}} \mathcal{L}_{\text{cover}}, \\ \mathcal{L} &= \mathcal{L}_{\text{det}} + \lambda_{\text{sel}} \mathcal{L}_{\text{sel}}, \end{aligned} \tag{8}$$

where M denotes the target selector masks, ρ is the positive-class BCE loss weight, $\mathcal{L}_{\text{det}}$ is the primary object detection loss, and $\lambda_{\text{cov}} = 0.5$, $\lambda_{\text{sel}} = 0.2$ are balancing hyperparameters.

3.3. Efficient Decoupled Head and Scale-Aware Loss

In high-resolution detection, standard decoupled detection heads (e.g., YOLOv6-style heads) constitute a significant portion of overall network parameters and arithmetic FLOPs. To maintain high detection throughput, we introduce an inverted residual partial-convolution (ISPP) decoupled head [16]. Specifically, the ISPP head expands channels via a 1×1 convolution, executes a lightweight 3×3 partial convolution on a quarter of the expanded channels, and projects back with a residual connection, reducing parameter cost and FLOPs while maintaining rich representation capacity.

Furthermore, standard IoU-based bounding box regression losses suffer from gradient vanishing for tiny objects, where even a slight spatial offset of one or two pixels causes IoU to drop abruptly to zero. To address this, we integrate the Scale-Aware Balancing Loss (SABL) [9].

As shown in Fig. 3, let $s = \sqrt{w_{\text{gt}} h_{\text{gt}}}$ be the geometric scale of a ground-truth box in input pixels, and let D_W denote the Normalized Gaussian Wasserstein distance between the predicted and ground-truth Gaussian box models $(c_x, c_y, w/2, h/2)$. A scale gating function $\mu(s)$ is formulated as:

$$\mu(s) = \exp\left(-\left(\frac{s}{\kappa}\right)^{\beta}\right), \quad \kappa = 32,\ \beta = 6. \tag{9}$$

The total bounding-box regression loss $\mathcal{L}_{\text{box}}$ smoothly interpolates between Wasserstein distance and Complete IoU

(CIoU):

$$\mathcal{L}_{\text{box}} = 1 - \text{IoU} + \mu(s)\Big(1 - e^{-D_W/C}\Big) + (1-\mu(s))\ell_{\text{ctr}} + \alpha v, \tag{10}$$

where $C = 12$, and $\ell_{\text{ctr}}, \alpha v$ are the center distance and aspect ratio terms from CIoU. When $s \to 0$, $\mu(s) \to 1$, allowing the Wasserstein metric to provide continuous, smooth gradients even under zero spatial IoU overlap.

4. EXPERIMENTAL SETUP

4.1. Datasets and Evaluation Protocols

We evaluate DES-ESOD on two representative, highly demanding small-object detection benchmarks spanning urban and maritime remote sensing domains:

UAVDT [2]: Captured by UAV platforms across varied urban traffic scenarios, UAVDT contains diverse viewing angles, altitudes, and intense scale variations. We evaluate on the official test split across three object classes (*car*, *truck*, and *bus*) under an input resolution of 1280×1280 pixels.

SeaPerson (TinyPersonV2) [4,10]: A challenging benchmark dedicated to tiny person detection in open water and coastal search-and-rescue contexts. The official test split contains 300,375 annotated instances across 5,752 images, with over 85% of instances falling into the *Very Tiny* and *Tiny* categories ($< 32 \times 32$ pixels). In accordance with realistic high-resolution deployment, we process full whole-frame images at 2048×2048 resolution without artificial pre-tiling.

4.2. Implementation Details

All models are implemented in PyTorch and utilize a standard YOLOv5m architecture as the base detection backbone for fair comparison. Models are trained for 50 epochs using the Stochastic Gradient Descent (SGD) optimizer with momentum 0.937, weight decay 5×10^{-4}, and an initial learning rate of 0.01 decayed via a cosine annealing schedule. Global batch size is set to 8. Spatial routing is performed using the dual-evidence priority map with fixed thresholding $\tau = 0.5$ (and $\tau = 0.3$ on UAVDT, matching baseline protocols).

4.3. Evaluation Metrics

Model accuracy is evaluated using standard COCO-style metrics, including mean Average Precision at IoU threshold 0.50 (mAP@.5) and averaged over IoU thresholds $0.50 : 0.05 : 0.95$ (mAP@.5 : .95). To rigorously evaluate tiny-object detection, we report size-bucketed detector recall at IoU 0.50 across four area scales: *Very Tiny* ($< 16 \times 16$ px), *Tiny* ($16 \times 16 - 32 \times 32$ px), *Small* ($32 \times 32 - 96 \times 96$ px), and *Medium/Large* ($> 96 \times 96$ px). Computational efficiency is measured via total GFLOPs, frame rate (FPS), per-image latency (ms) on an NVIDIA GPU at batch size 1, and the spatial area retention ratio (Occupy).

Table 2. Quantitative performance comparison on the UAVDT test set.

Method	mAP@.5	mAP@.5:.95	Recall	VT-Recall	Occupy
ESOD Baseline [3]	0.344	0.171	84.67%	83.63%	0.245
DES-ESOD (Ours)	**0.371**	**0.195**	**90.17%**	**84.20%**	0.480
Improvement (Δ)	**+2.7 pp**	**+2.4 pp**	**+5.5 pp**	**+0.57 pp**	–

Table 3. Detection accuracy and runtime efficiency on SeaPerson.

Method	mAP@.5	mAP@.5:.95	Recall	BPR	GFLOPs	FPS	Occupy
ESOD Baseline [3]	0.750	0.320	69.6%	0.947	202.4	88.9	0.228
+ Soft-Coverage	**0.769**	**0.325**	**70.8%**	**0.991**	266.8	75.4	0.363
Improvement (Δ)	**+1.9 pp**	**+0.5 pp**	**+1.2 pp**	**+0.044**	–	–	–

5. RESULTS AND ANALYSIS

5.1. Quantitative Results on UAVDT and SeaPerson

Table 2 presents the primary detection results on the UAVDT benchmark. DES-ESOD demonstrates consistent and substantial improvements across all key evaluation metrics compared to the standard ESOD baseline. In particular, DES-ESOD achieves a **+2.7 pp** increase in mAP@.5 (0.344 vs. 0.371), a **+2.4 pp** increase in mAP@.5 : .95 (0.171 vs. 0.195), and a remarkable **+5.5 pp** boost in total detector recall (84.67% vs. 90.17%). Per-class recall consistently improves across all vehicle categories: *car* ($84.87\% \to 90.56\%$, **+5.69 pp**), *truck* ($84.41\% \to 84.88\%$, **+0.47 pp**), and *bus* ($75.20\% \to 76.28\%$, **+1.08 pp**).

Table 3 evaluates performance on the dense SeaPerson maritime dataset. Incorporating object-level soft-coverage loss into the spatial selector directly improves mAP@.5 by **+1.9 pp** ($0.750 \to 0.769$) and overall recall by **+1.2 pp**, while increasing Bounding Patch Recall (BPR) from 0.947 to 0.991. The system sustains a high frame rate of 75.4 FPS at 2048×2048 resolution, confirming real-time execution capability.

5.2. Size-Stratified Recall Analysis across Scale Strata

To rigorously evaluate how our method benefits objects at different physical scales, Table 4 provides a comprehensive size-stratified breakdown of detector recall across all 300,375 test instances in SeaPerson. Targets are partitioned into four standard physical area bins: *Very Tiny* ($< 16 \times 16$ px, 82,417 targets), *Tiny* ($16 \times 16 - 32 \times 32$ px, 174,816 targets), *Small* ($32 \times 32 - 96 \times 96$ px, 42,987 targets), and *Medium/Large* ($> 96 \times 96$ px, 155 targets).

As shown in Table 4, coverage-guided spatial selection delivers decisive recall improvements across every scale bucket. Crucially, the largest absolute recall gain occurs

Table 4. Detailed size-bucket recall comparison across 300,375 ground-truth test instances on SeaPerson.

Scale Bucket	Pixel Area	Baseline [3]	+ Soft-Coverage	Gain (Δ)	Relative
Very Tiny	$< 16^2$ px	74.03%	**75.49%**	**+1.46 pp**	+1.97%
Tiny	$16^2 - 32^2$ px	86.78%	**91.57%**	**+4.79 pp**	+5.52%
Small	$32^2 - 96^2$ px	94.74%	**95.71%**	**+0.97 pp**	+1.02%
Med/Large	$> 96^2$ px	79.35%	**81.94%**	**+2.59 pp**	+3.26%
Overall	**All Scales**	**84.42%**	87.75%	**+3.33 pp**	**+3.94%**

Table 5. Diagnostic breakdown of missed Very Tiny instances ($< 16^2$ px).

Method	Missed Targets	Selector Dropped	Localization Failure	Other
ESOD Baseline	21,388	25.6%	74.3%	0.1%
+ Soft-Coverage	**20,181**	**20.1%**	79.7%	0.1%
Impact (Δ)	**-1,207**	**-5.5 pp**	+5.4 pp	0.0%

in the dominant *Tiny* stratum, surging by **+4.79 pp** (from 86.78% to 91.57%, representing over 8,380 additional correctly recovered targets). For the most challenging *Very Tiny* targets, recall improves by **+1.46 pp** (74.03% $\rightarrow$ 75.49%, +1,203 recovered instances). In total, detector recall jumps from 84.42% to 87.75% (+3.33 pp overall), demonstrating that soft-coverage optimization directly enhances sensitivity where semantic signals are weakest.

5.3. Error Attribution and Recall Safety Diagnostics

To isolate the precise mechanism contributing to the recall gain, Table 5 presents an in-depth error attribution diagnostic on all missed *Very Tiny* instances ($< 16 \times 16$ px). Missed targets are categorized into: (i) *Selector Dropped* (no candidate patch retained near the target); and (ii) *Localization Failure* (candidate patch retained, but detection head prediction has IoU < 0.5).

The baseline ESOD model suffered a high selector drop rate of 25.6% on missed tiny targets due to semantic activation collapse. In contrast, training with the proposed object-level soft-coverage loss reduces the selector-dropped error rate to **20.1%** (**-5.5 pp**) while eliminating 1,207 missed instances. This confirms that soft-coverage supervision successfully rectifies early spatial pruning errors, effectively shifting the primary remaining detection bottleneck toward downstream localization refinement (which is addressed by SABL and ISPP decoupled heads).

6. CONCLUSION

In this paper, we presented DES-ESOD, an efficient dual-evidence selection framework tailored for recall-safe, high-resolution small object detection in aerial and maritime vision. By addressing the critical vulnerability of semantic feature collapse in single-evidence selective computation, DES-ESOD introduces a dual-evidence priority head that combines semantic objectness with channel-pooled spectral saliency at negligible compute overhead ($\mathcal{O}(1)$ relative to backbone width). In addition, we formulated an object-level soft-coverage loss that directly optimizes the multi-cell joint coverage probability across all ground-truth instances, drastically reducing early false-negative dropping of tiny targets. Paired with a lightweight ISPP-based decoupled detection head and scale-aware bounding-box regression (SABL), DES-ESOD achieves substantial improvements in both detection accuracy and tiny-object recall across challenging benchmarks including UAVDT and SeaPerson, while maintaining real-time inference speeds. Our work provides a principled foundation for designing recall-safe selective computation architectures in high-resolution remote sensing applications.

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